



King Saud University
College of Computer and Information Sciences
Department of Information Technology

IT222: Database Principles
2nd Semester 1447 H



مستشفى د. سليمان فقيه
Dr. Soliman Fakeeh Hospital

Dr. Soliman Fakeeh Hospital

Phase # 3

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1. Project Description:

A database application that is used by staff at Dr. Soliman Fakeeh Hospital to manage hospital data and daily operations. It stores and updates information about patients, doctors, departments, and appointments.

The system is used to register patients, schedule appointments, record diagnoses and treatments, manage admissions and discharges, process billing and insurance data, and track laboratory and pharmacy services. It also manages room and bed assignments and generates basic operational reports.

2. View Description:

This application is intended for hospital administrative and medical staff at Dr. Soliman Fakeeh Hospital. Users of this view include receptionists, nurses, and doctors who are responsible for handling patient-related tasks and daily hospital operations.

Their responsibilities include registering patients, updating medical records, managing appointments, and monitoring admissions and discharges. The data they are interested in includes patient personal details, visit history, assigned doctors, treatments, and appointment schedules.

3. Data Requirements:

Doctor

Stores information about medical practitioners in the hospital. It includes a unique DoctorID (PK) to identify each doctor, along with the doctor's Name, medical Specialization, assigned ClinicRoomNum, and clinic Extension number.

- Related to Appointment (Multiplicity 1:*) : A doctor can have multiple appointments.
- Related to Department (Multiplicity *:1): Each doctor belongs to one department.

Medicine

Stores information about pharmaceutical products available in the hospital pharmacy. It includes a unique CommercialName (PK), the Price of the medicine, its ExpiryDate, and the available Quantity in inventory.

- Related to Doctor (Multiplicity * : *) : A medicine can be prescribed by multiple doctors.

- Related to Patient (Multiplicity * : *): A medicine can be given to multiple patients.

Patient

Stores information about registered patients in the hospital system. It includes a unique FileNumber (PK) for identification, the patient's NationalID, DateOfBirth, Gender, and BloodType for medical and administrative purposes.

- With Appointment: The relationship is One-to-Many (1: *).
- With Medicine :The relationship is Many-to-Many (* : *).

Appointment

Stores information related to patient visits and scheduling. It includes a unique AppID (PK), the AppointmentDate, AppointmentTime, the appointment Status, and the VisitType indicating the nature of the visit.

- Related to Patient (Multiplicity *:1) Each appointment belongs to one patient.
- Related to Doctor (Multiplicity: *:1) Each appointment is assigned to one doctor.

Department

Stores information about hospital departments. It includes a unique DeptCode (PK), the DeptName, the department Location, and a ContactNo for official communication.

- With Doctor: The relationship is One-to-Many (1: *).
- With Appointment The relationship is One-to-Many (1: *)

4. Transaction Requirements:

Data Entry:

1. Register New Patient:

Add a new patient record to the system with their personal details (Name, ID, Phone, etc.).

2. Book New Appointment:

Insert a new appointment record linking a specific Patient with a Doctor at a chosen time.

Data update/deletion:

1. Cancel Appointment:

Update the status of an existing appointment to "Canceled" if the patient cannot make it.

2. Update Contact Info:

Modify the Phone Number of an existing patient in case it changes.

3. Update Medicine Stock:

Decrease the Stock Level of a medicine after it has been dispensed to a patient.

4. Delete Prescription Record:

Delete a prescription entry if it was entered incorrectly.

5. Update Doctor Department:

Update the department of a doctor if they are transferred to another department.

Data Queries:

1. Search for Patient:

Retrieve the full details (Name, Age, Blood Type) of a patient using their unique File Number.

2. View Daily Appointments:

List all scheduled appointments for a specific Doctor on a given Date.

3. Find Doctors by Department:

Display a list of all doctors working in a specific Department (e.g., Pediatrics).

4. Check Medicine Stock:

Check the current Stock Level of a specific Medicine to see if it is available in the pharmacy.

5. View Patient History:

Retrieve all past Appointments for a specific Patient to review their visit history.

6. Find Doctor Contact Info:

Get the Coordinator Name and Clinic Extension for a specific doctor.

7. Check Medicine Price:

Search for a Medicine by its commercial name to retrieve its Price.

8. List Department Details:

Retrieve the Location and Contact Number of a specific Department.

9. Filter Appointments by Status:

List all appointments that have the status "Confirmed" for a specific day.

10. Check Expiring Medicines:

List all medicines that have an Expiry Date within the current month.

5. Global enhanced entity relationship diagram (EER):

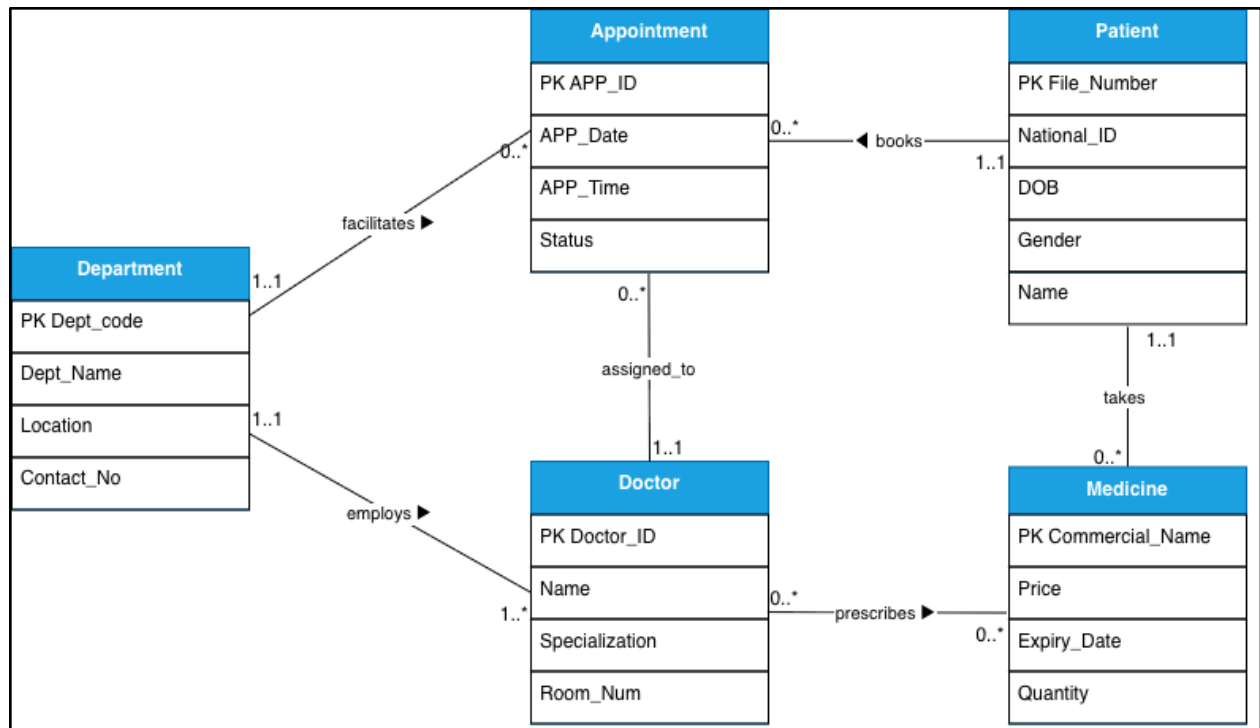


Figure 1. ERR Diagram

6. Relational Schema:

Doctor (DoctorID, Name, Specialization, RoomNum, Dept_code)

Primary key: DoctorID

Foreign key: Dept_code references Department (Dept_code)

Patient (FileNumber, NationalID, DOB, Gender)

Primary key: FileNumber

Foreign key: CommercialName references Medicine (CommercialName)

Department (Dept_code, Dept_Name, Location, ContactNo)

Primary key: Dept_code

Appointment (APP_ID, APP_Date, APP_Time, Status, Dept_code, FileNumber, DoctorID)

Primary key: APP_ID

Foreign key: DoctorID References Doctor (DoctorID)

Foreign key: FileNumber References Patient (FileNumber)

Foreign key: Dept_code references Department (Dept_code)

Prescribes (DoctorID, CommercialName)

Primary key: DoctorID and CommercialName

Foreign key: DoctorID References Doctor (DoctorID)

Foreign key: CommercialName References Medicine (CommercialName)

Medicine (CommercialName, Price, Expiry_Date, Quantity)

Primary key: CommercialName

Takes (FileNumber, CommercialName)

Primary key: FileNumber and CommercialName

Foreign key: FileNumber References Patient (FileNumber)

Foreign key: CommercialName References Medicine (CommercialName)

7. Data Dictionary showing description of all entities:

Entity Name	Description	Occurrence
Doctor	Stores information about medical practitioners including specialization and clinic details	each doctor can be assigned to one, none or many appointments and prescribe one, none or many medicines
Medicine	Stores information about pharmaceutical products available in the hospital pharmacy	Each medicine can be taken by one,none or many patients
Patient	Stores information about registered patients including personal and medical details	A patient can book one, none or many appointments and can receive one , none or many medicines
Appointment	Stores scheduling information linking patients with doctors for visits	Each appointment belongs to one patient and is assigned to one doctor
Department	Stores information about hospital departments including location and contact	Each department employs one or more doctor and facilitates one or more appointment

Table 1. Data Dictionary of entities

8. Data Dictionary showing description of all relationships:

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
Department	1..1	employs	Doctor	1..*
Department	1..1	facilitates	Appointment	0..*
Patient	1..1	books	Appointment	0..*
Doctor	1..1	assigned_to	Appointment	0..*
Doctor	0..*	prescribes	Medicine	0..*
Patient	1..1	takes	Medicine	0..*

Table 2. Data Dictionary of relationships

9. Data Dictionary showing description of all attributes:

Entity Name	Attribute	Description	Data Type	Length	Nulls	Multi-Valued	Default Value	Range	PK
Doctor	+Doctor_ID +Name +Specialization +Room_Num	Stores information about medical practitioners in the hospital	Varchar Varchar Varchar Varchar	7 20 10 5	No No No No	No No No No	-	-	Doctor_ID
Medicine	+Commercial_Name +Price +Expiry_Date +Quantity	Stores information about pharmaceutical products available in the hospital pharmacy.	Varchar Decimal () Date int	25 10 3	No No No Yes	No No No No	-	-	Commercial_Name
Department	+Dept_code +Dept_Name +Location +Contact_No	Stores information about hospital departments	Varchar Varchar Varchar Int	7 25 10 10	No No No Yes	No No No Yes	-	-	Dept_code
Appointment	+APP_ID +APP_DATE +APP_Time +Status	Stores information related to patient visits and scheduling.	Varchar Date Int Varchar	15 10	No No No No	No No No No	-	-	APP_ID

[IT222-1447(2): Dr. Soliman Fakeeh Hospital]

Patient	+File_Number +National_ID +DOB +Gender	Stores information about registered patients in the hospital system	Varchar Int Date Char	7 10 1	No No No No	No No No No	-	F,M	File_Num ber
---------	---	---	--------------------------------	------------------	----------------------	----------------------	---	-----	-----------------

Table 3. Data Dictionary of attributes

10.DB tables creation:

Table *Appointment*:

The screenshot shows a database management interface with a sidebar on the left containing 'Object Browser' and a search bar. The main area displays the 'APPOINTMENT' table structure. The 'DDL' tab is selected, showing the following SQL code:

```

1 CREATE TABLE "APPOINTMENT"
2 (
3   "APP_ID" NUMBER,
4   "APP_DATE" DATE NOT NULL ENABLE,
5   "APP_TIME" VARCHAR2(10),
6   "STATUS" VARCHAR2(20),
7   "DEPT_CODE" VARCHAR2(10) NOT NULL ENABLE,
8   "FILENUMBER" NUMBER NOT NULL ENABLE,
9   "DOCTORID" NUMBER NOT NULL ENABLE,
10  PRIMARY KEY ("APP_ID")
11  USING INDEX ENABLE
12 ) ;
  
```

The interface also includes tabs for 'Columns', 'Data', 'Indexes', 'Constraints', 'Grants', 'Statistics', 'Triggers', 'Dependencies', and 'Sample Queries'. The 'Schema' dropdown is set to 'WKSP_DBPROJECTSTUDEN'.

Object Browser

Schema WKSP_DBPROJECTSTUDEN

Type to filter...

Tables

APPOINTMENT

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MEDICINE

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APPOINTMENT

ColumnsDataIndexesConstraintsGrantsStatisticsTriggersDependenciesDDLSample Queries

+ Add ColumnModify ColumnRename ColumnDrop ColumnUI DefaultsRefreshMore

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
APP_ID	NUMBER	N		1		
APP_DATE	DATE	N				
APP_TIME	VARCHAR2(10 BYTE)	Y				
STATUS	VARCHAR2(20 BYTE)	Y				
DEPT_CODE	VARCHAR2(10 BYTE)	N				
FILENUMBER	NUMBER	N				
DOCTORID	NUMBER	N				

1 cells selected

1 - 7 of 7

danalturaff@gmail.comen

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Table *Department*:

Object Browser

Schema WKSP_DBPROJECTSTUDEN

Type to filter...

Tables

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DEPARTMENT

DOCTOR

MEDICINE

PATIENT

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DEPARTMENT

ColumnsDataIndexesConstraintsGrantsStatisticsTriggersDependenciesDDLSample Queries

DownloadRefresh

1

2

3

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5

6

7

8

CREATE TABLE "DEPARTMENT"

{

"DEPT_CODE" VARCHAR2(10),

"DEPT_NAME" VARCHAR2(50) NOT NULL ENABLE,

"LOCATION" VARCHAR2(50),

"CONTACTING" NUMBER(10,0) NOT NULL ENABLE,

PRIMARY KEY ("DEPT_CODE")

USING INDEX ENABLE

}

;

↑ Object Browser

Type to filter...

+

▼

Tables

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Schema WKSP_DBPROJECTSTUDEN

DEPARTMENT

ColumnsDataIndexesConstraintsGrantsStatisticsTriggersDependenciesDDLSample Queries

+ Add Column✎ Modify Column📄 Rename Column🗑 Drop Column🔄 UI Defaults🔄 RefreshMore

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
DEPT_CODE	VARCHAR2(10 BYTE)	N		1		
DEPT_NAME	VARCHAR2(50 BYTE)	N				
LOCATION	VARCHAR2(50 BYTE)	Y				
CONTACTNO	NUMBER(10,0)	N				

1 cells selected1 - 4 of 4

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Table *Doctor*:

↑ Object Browser

Type to filter...

+

▼

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Schema WKSP_DBPROJECTSTUDEN

DOCTOR

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Download🔄 Refresh

🔄🔄🔍A-Z

1

CREATE TABLE "DOCTOR"

2

{

3

"DOCTORID" NUMBER,

4

"NAME" VARCHAR2(50) NOT NULL ENABLE,

5

"SPECIALIZATION" VARCHAR2(50),

6

"ROOMNUM" NUMBER,

7

"DEPT_CODE" VARCHAR2(10) NOT NULL ENABLE,

8

PRIMARY KEY ("DOCTORID")

9

} ;

10

USING INDEX ENABLE

11

);

12

ALTER TABLE "DOCTOR" ADD CONSTRAINT "FK_DOC_DEPT" FOREIGN KEY ("DEPT_CODE")

REFERENCES "DEPARTMENT" ("DEPT_CODE") ENABLE;

Object Browser

Schema: WKSP_DBPROJECTSTUDEN

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DOCTOR

Columns

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
DOCTORID	NUMBER	N		1		
NAME	VARCHAR2(50 BYTE)	Y				
SPECIALIZATION	VARCHAR2(50 BYTE)	Y				
ROOMNUM	NUMBER	Y				
DEPT_CODE	VARCHAR2(10 BYTE)	Y				

1 cells selected

1 - 5 of 5

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Table *Medicine*:

Object Browser

Schema: WKSP_DBPROJECTSTUDEN

Tables

- APPOINTMENT
- DEPARTMENT
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```
1 CREATE TABLE "MEDICINE"
2 (
3   "COMMERCIALNAME" VARCHAR2(50),
4   "PRICE" NUMBER(10,2),
5   "EXPIRY_DATE" DATE,
6   "QUANTITY" NUMBER,
7   PRIMARY KEY ("COMMERCIALNAME")
8   USING INDEX ENABLE
9 );
```

1 cells selected

1 - 4 of 4

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Object Browser Schema: WKSP_DBPROJECTSTUDEN

Search: Type to filter...

Tables:

- APPOINTMENT
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Columns Data Indexes Constraints Grants Statistics Triggers Dependencies DDL Sample Queries

+ Add Column Modify Column Rename Column Drop Column UI Defaults Refresh More

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
COMMERCIALNAME	VARCHAR2(50 BYTE)	N		1		
PRICE	NUMBER(10,2)	Y				
EXPIRY_DATE	DATE	Y				
QUANTITY	NUMBER	Y				

1 cells selected 1 - 4 of 4

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Table *Patient*:

Object Browser Schema: WKSP_DBPROJECTSTUDEN

Search: Type to filter...

Tables:

- APPOINTMENT
- DEPARTMENT
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- PRESCRIBES
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```
1 CREATE TABLE "PATIENT"
2 (
3   "FILENUMBER" NUMBER,
4   "NATIONALID" NUMBER(10,0) NOT NULL ENABLE,
5   "DOB" DATE NOT NULL ENABLE,
6   "GENDER" CHAR(1) NOT NULL ENABLE,
7   "NAME" VARCHAR2(25) NOT NULL ENABLE,
8   PRIMARY KEY ("FILENUMBER")
9   USING INDEX ENABLE
10 );
```

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Object Browser

Schema: WKSP_DBPROJECTSTUDEN

Search: Type to filter...

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+ Add Column | Modify Column | Rename Column | Drop Column | UI Defaults | Refresh | More

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
FILENUMBER	NUMBER	N		1		
NATIONALID	NUMBER(10,0)	N				
DOB	DATE	N				
GENDER	CHAR(1 BYTE)	N				
NAME	VARCHAR2(25 BYTE)	N				

1 cells selected

1 - 5 of 5

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Table *Prescribes*:

Object Browser

Schema: WKSP_DBPROJECTSTUDEN

Search: Type to filter...

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- APPOINTMENT
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```
1 CREATE TABLE "PRESCRIBES"
2 (
3   "DOCTORID" NUMBER,
4   "COMMERCIALNAME" VARCHAR2(50),
5   CONSTRAINT "PK_PRE" PRIMARY KEY ("DOCTORID", "COMMERCIALNAME")
6   USING INDEX ENABLE
7 );
8
9 ALTER TABLE "PRESCRIBES" ADD CONSTRAINT "FK_PRE_DOC" FOREIGN KEY ("DOCTORID")
10 REFERENCES "DOCTOR" ("DOCTORID") ENABLE;
11 ALTER TABLE "PRESCRIBES" ADD CONSTRAINT "FK_PRE_MED" FOREIGN KEY ("COMMERCIALNAME")
12 REFERENCES "MEDICINE" ("COMMERCIALNAME") ENABLE;
```

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Object Browser

Schema: WKSP_DBPROJECTSTUDEN

Search: Type to filter...

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+ Add Column | Modify Column | Rename Column | Drop Column | UI Defaults | Refresh | More

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
DOCTORID	NUMBER	N		1		
COMMERCIALNAME	VARCHAR2(50 BYTE)	N		2		

Table *Takes*:

Object Browser

↑

Type to filter...

+

▼

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11

CREATE TABLE "TAKES"

{

"FILENUMBER" NUMBER,

"COMMERCIALNAME" VARCHAR2(50),

CONSTRAINT "PK_TAKES" PRIMARY KEY ("FILENUMBER", "COMMERCIALNAME")

USING INDEX ENABLE

};

ALTER TABLE "TAKES" ADD CONSTRAINT "FK_TAKES_MED" FOREIGN KEY ("COMMERCIALNAME")

REFERENCES "MEDICINE" ("COMMERCIALNAME") ENABLE;

ALTER TABLE "TAKES" ADD CONSTRAINT "FK_TAKES_PATIENT" FOREIGN KEY ("FILENUMBER")

REFERENCES "PATIENT" ("FILENUMBER") ENABLE;

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↑

Type to filter...

+

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+ Add Column

✎ Modify Column

🏷️ Rename Column

🗑️ Drop Column

⚙️ UI Defaults

🔄 Refresh

More ▼

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
FILENUMBER	NUMBER	N		1		
COMMERCIALNAME	VARCHAR2(50 BYTE)	N		2		

danaluraifi@gmail.com

en

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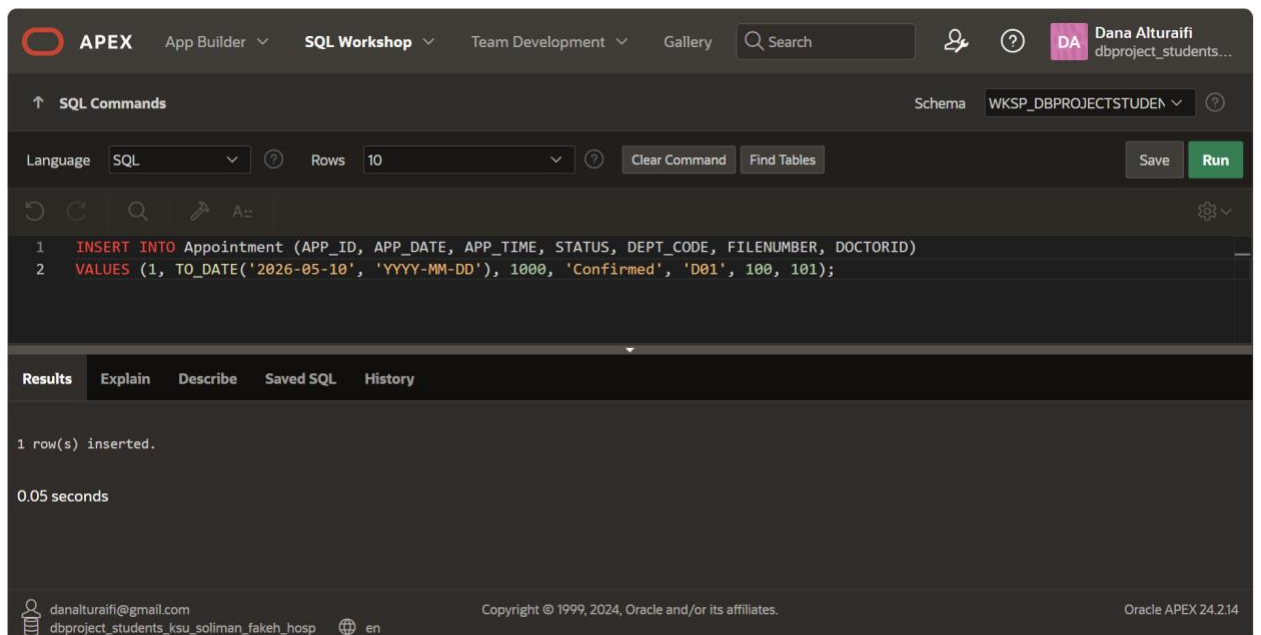
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11.Data insertion:

Insert Table Appointment:

Command:



The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is 'Dana Alturaifi' with the role 'dbproject_students...'. The 'SQL Commands' tab is active, showing a schema of 'WKSP_DBPROJECTSTUDEN'. The language is set to 'SQL' and the number of rows is '10'. The SQL command entered is:

```
1 INSERT INTO Appointment (APP_ID, APP_DATE, APP_TIME, STATUS, DEPT_CODE, FILENUMBER, DOCTORID)
2 VALUES (1, TO_DATE('2026-05-10', 'YYYY-MM-DD'), 1000, 'Confirmed', 'D01', 100, 101);
```

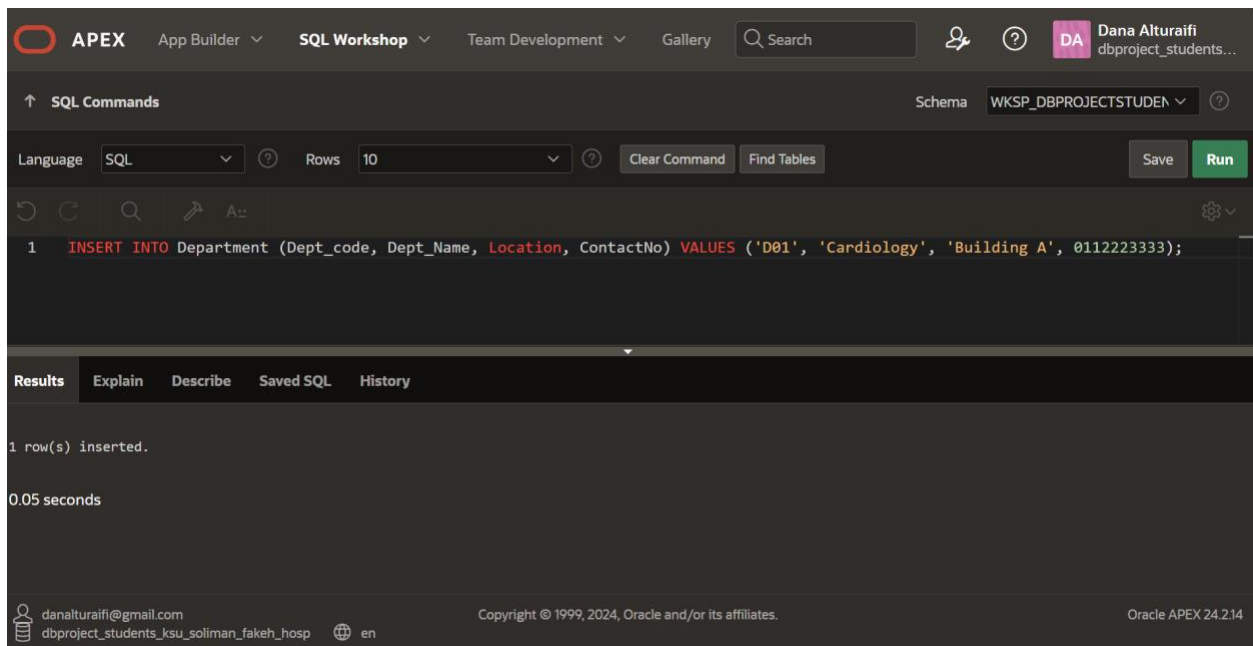
The 'Results' tab shows '1 row(s) inserted.' and '0.05 seconds'.

Result:

APPOINTMENT							
Columns Data Indexes Constraints Grants Statistics Triggers Dependencies DDL Sample Queries							
+ Insert Row Columns... Filter... Count Rows Load Data Download Refresh							
	APP_ID	APP_DATE	APP_TIME	STATUS	DEPT_CODE	FILENUMBER	DOCTORID
	3	5/20/2026	930	Pending	D01	300	103
	2	5/15/2026	1100	Confirmed	D01	200	102
	103	5/25/2026	1400	Confirmed	D02	500	201
	1	5/10/2026	1000	Confirmed	D01	100	101
	50	6/1/2026	900	Confirmed	D01	500	201
	101	5/20/2026	900	Confirmed	D02	500	201

Insert Table Department:

Command:



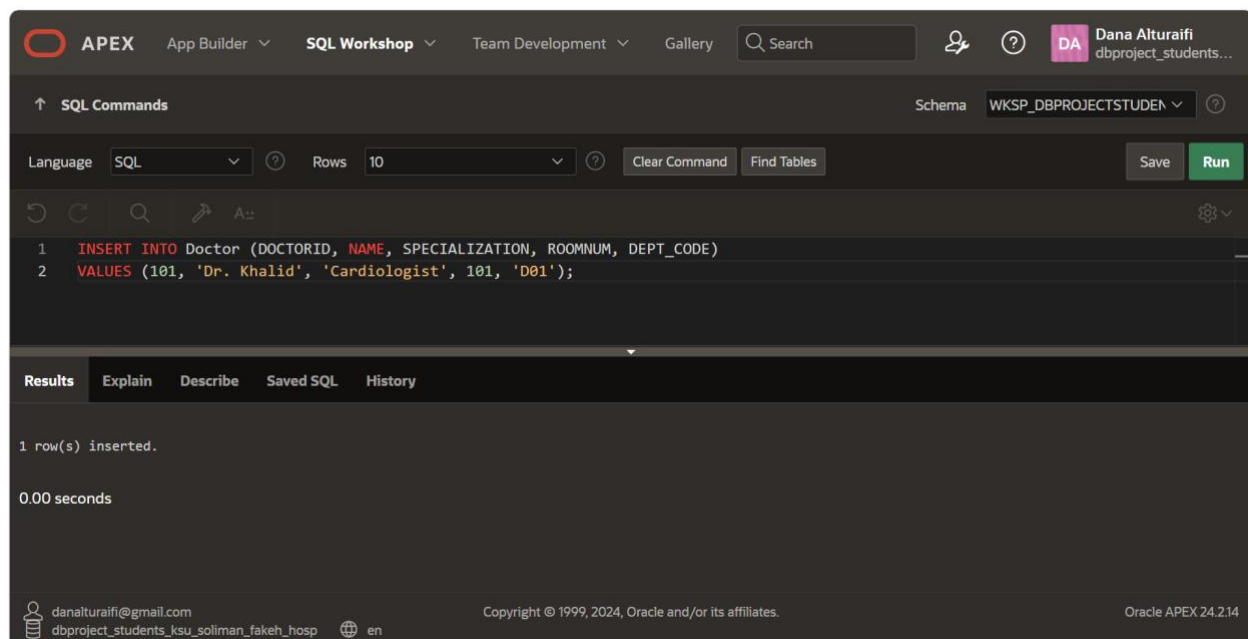
The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is logged in as 'Dana Alturaifi' with the schema 'WKSP_DBPROJECTSTUDEN'. The 'SQL Commands' tab is active, showing a single SQL command: `INSERT INTO Department (Dept_code, Dept_Name, Location, ContactNo) VALUES ('D01', 'Cardiology', 'Building A', 0112223333);`. The command is executed, resulting in '1 row(s) inserted.' and a duration of '0.05 seconds'. The bottom status bar shows the user's email 'danalturaifi@gmail.com', the database name 'dbproject_students_ksu_soliman_fakeh_hosp', and the Oracle APEX version '24.2.14'.

Result:

DEPARTMENT				
Columns Data Indexes Constraints Grants Statistics Triggers Dependencies DDL Sample Queries				
+ Insert Row Columns... Filter... Count Rows Load Data Download Refresh				
	DEPT_CODE	DEPT_NAME	LOCATION	CONTACTNO
	D02	Radiology	Building B	112224444
	D01	Cardiology	Building A	112223333
	D03	Laboratory	Building C	112225555
	D04	Emergency	Building A	112226666

Insert Table Doctor:

Command :

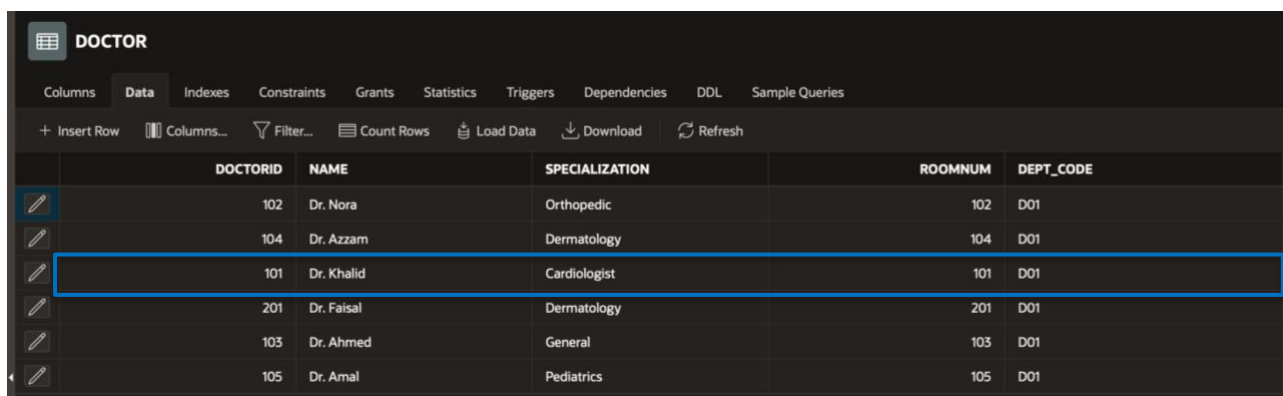


The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is logged in as 'Dana Altaifi' with the session 'dbproject_students...'. The 'SQL Commands' tab is active, showing a schema of 'WKSP_DBPROJECTSTUDEN'. The language is set to 'SQL' and the number of rows is '10'. The SQL command entered is:
1 INSERT INTO Doctor (DOCTORID, NAME, SPECIALIZATION, ROOMNUM, DEPT_CODE)
2 VALUES (101, 'Dr. Khalid', 'Cardiologist', 101, 'D01');
The 'Results' tab shows '1 row(s) inserted.' and '0.00 seconds'.

```
1 INSERT INTO Doctor (DOCTORID, NAME, SPECIALIZATION, ROOMNUM, DEPT_CODE)
2 VALUES (101, 'Dr. Khalid', 'Cardiologist', 101, 'D01');
```

Results: 1 row(s) inserted. 0.00 seconds

Result:

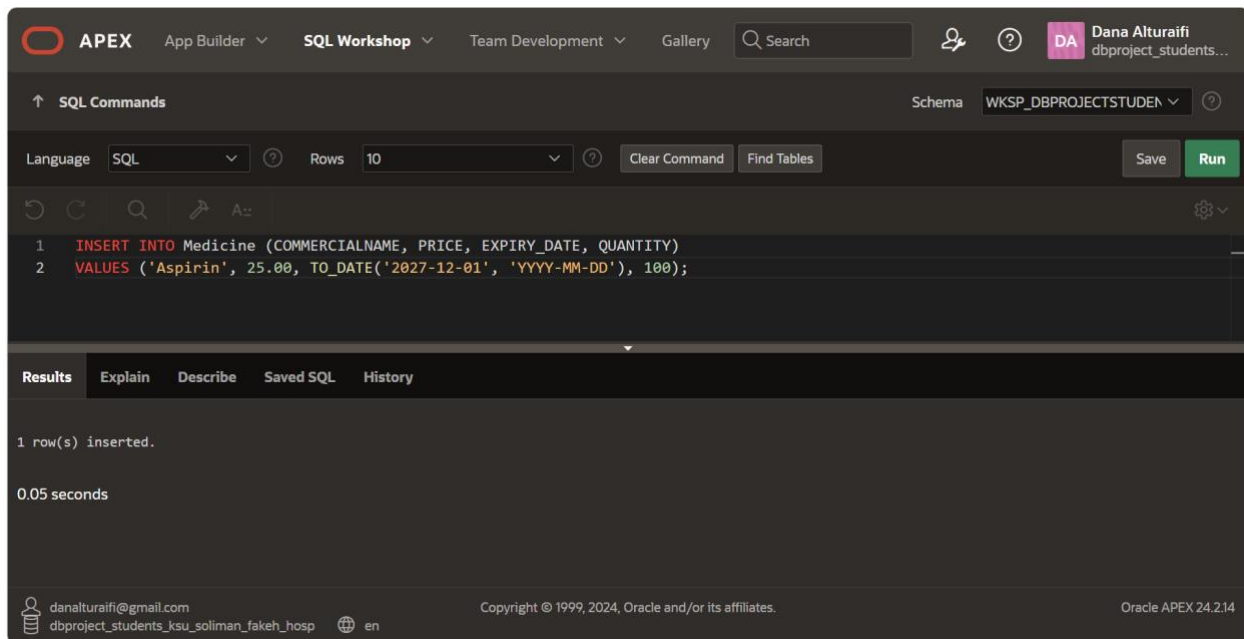


The screenshot shows the 'DOCTOR' table in the Oracle APEX interface. The table has columns: DOCTORID, NAME, SPECIALIZATION, ROOMNUM, and DEPT_CODE. The row with DOCTORID 101, NAME 'Dr. Khalid', SPECIALIZATION 'Cardiologist', ROOMNUM 101, and DEPT_CODE 'D01' is highlighted with a blue border.

DOCTORID	NAME	SPECIALIZATION	ROOMNUM	DEPT_CODE
102	Dr. Nora	Orthopedic	102	D01
104	Dr. Azzam	Dermatology	104	D01
101	Dr. Khalid	Cardiologist	101	D01
201	Dr. Faisal	Dermatology	201	D01
103	Dr. Ahmed	General	103	D01
105	Dr. Amal	Pediatrics	105	D01

Insert Table Medicine:

Command :



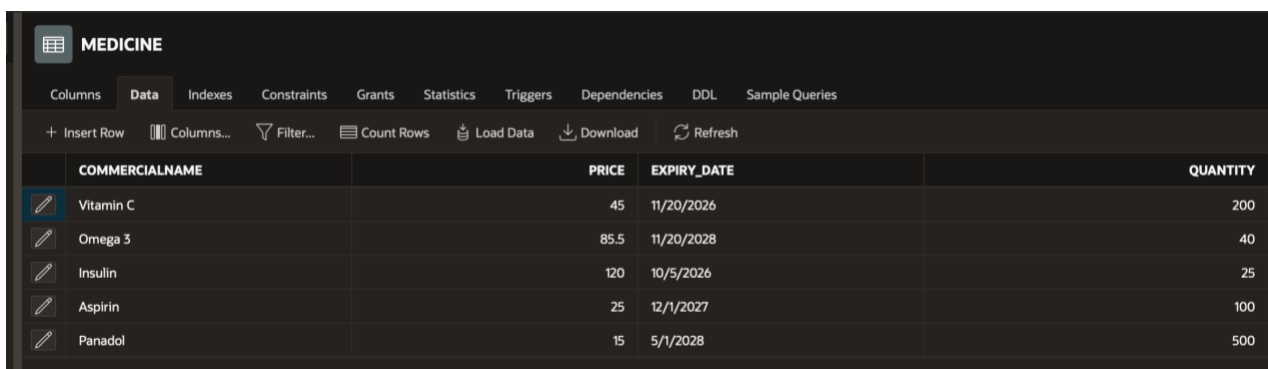
The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is logged in as 'Dana Alturaifi' with the email 'dbproject_students...'. The 'SQL Commands' section is active, showing a schema of 'WKSP_DBPROJECTSTUDEN'. The command entered is:

```
1 INSERT INTO Medicine (COMMERCIALNAME, PRICE, EXPIRY_DATE, QUANTITY)
2 VALUES ('Aspirin', 25.00, TO_DATE('2027-12-01', 'YYYY-MM-DD'), 100);
```

The 'Results' tab shows '1 row(s) inserted.' and '0.05 seconds'.

Footer information includes the email 'danalturaifi@gmail.com', the project name 'dbproject_students_ksu_soliman_fakeh_hosp', and the version 'Oracle APEX 24.2.14'.

Result:

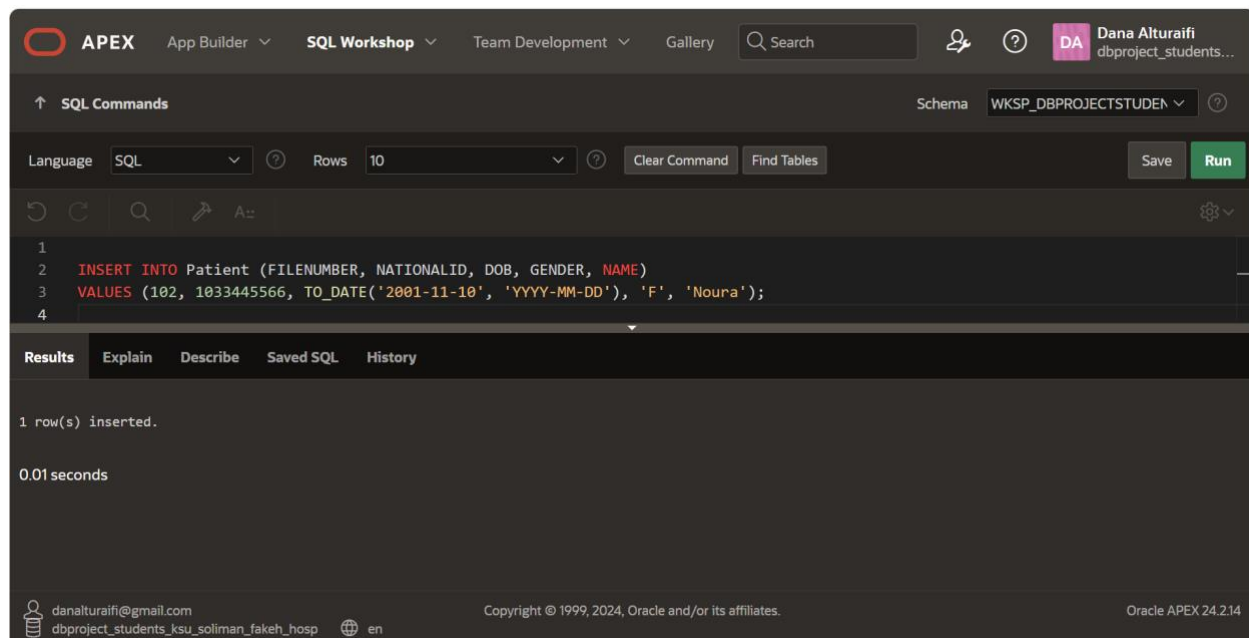


The screenshot shows the 'MEDICINE' table in the Oracle APEX interface. The table has five columns: 'COMMERCIALNAME', 'PRICE', 'EXPIRY_DATE', and 'QUANTITY'. The data is as follows:

	COMMERCIALNAME	PRICE	EXPIRY_DATE	QUANTITY
	Vitamin C	45	11/20/2026	200
	Omega 3	85.5	11/20/2028	40
	Insulin	120	10/5/2026	25
	Aspirin	25	12/1/2027	100
	Panadol	15	5/1/2028	500

Insert Table Patient:

Command:



The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes the APEX logo, App Builder, SQL Workshop, Team Development, and Gallery. A search bar and user profile (Dana Alturaifi) are also present. The main area is titled 'SQL Commands' and shows the 'Schema' as 'WKSP_DBPROJECTSTUDEN'. The 'Language' is set to 'SQL' and 'Rows' to '10'. The SQL command being executed is:

```
1  
2 INSERT INTO Patient (FILENUMBER, NATIONALID, DOB, GENDER, NAME)  
3 VALUES (102, 1033445566, TO_DATE('2001-11-10', 'YYYY-MM-DD'), 'F', 'Noura');  
4
```

The 'Results' tab is selected, showing '1 row(s) inserted.' and '0.01 seconds'. The bottom of the interface displays the user's email (danalturaifi@gmail.com), the database name (dbproject_students_ksu_soliman_fakeh_hosp), and the Oracle APEX version (24.2.14).

Result:

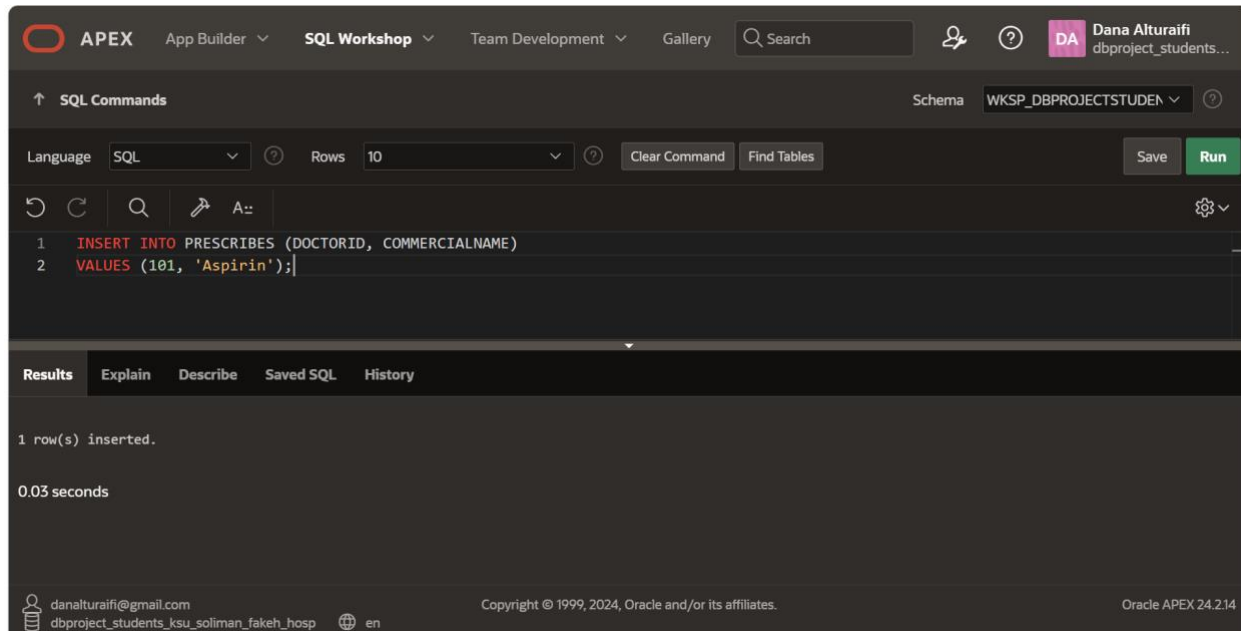
[IT222-1447(2): Dr. Soliman Fakeeh Hospital]

PATIENT					
Columns Data Indexes Constraints Grants Statistics Triggers Dependencies DDL Sample Queries					
+ Insert Row Columns... Filter... Count Rows Load Data Download Refresh					
	FILENUMBER	NATIONALID	DOB	GENDER	NAME
	200	1088776655	10/15/1995	M	Fahad
	102	1033445566	11/10/2001	F	Noura
	100	1099887766	5/20/2004	F	Layan
	300	1077665544	1/12/1990	F	Sara
	500	1122334499	3/10/1997	M	Sultan

[IT222-1447(2): Dr. Soliman Fakeeh Hospital]

Insert Table Prescribes:

Command:

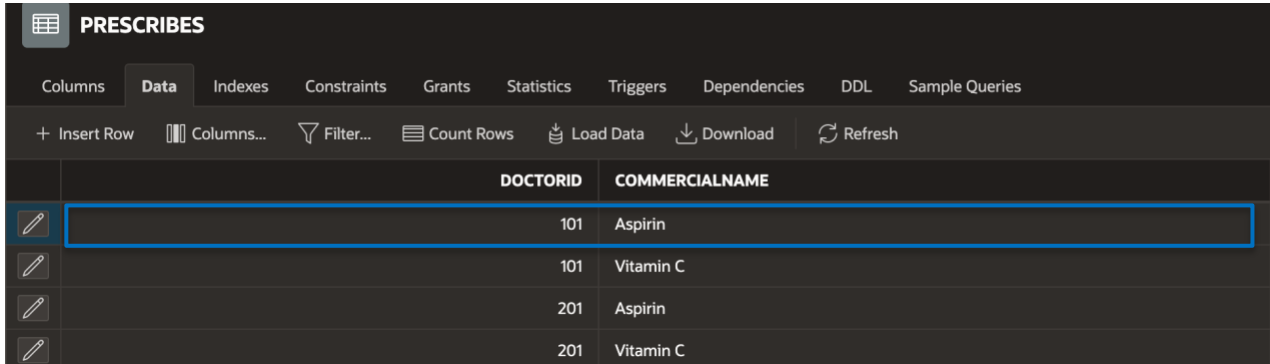


The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is logged in as 'Dana Alturaifi' with the role 'dbproject_students...'. The 'SQL Commands' panel is active, showing a schema of 'WKSP_DBPROJECTSTUDEN'. The command entered is:

```
1 INSERT INTO PRESCRIBES (DOCTORID, COMMERCIALNAME)
2 VALUES (101, 'Aspirin');
```

The 'Results' tab shows the execution output: '1 row(s) inserted.' and '0.03 seconds'. The footer includes the user's email 'danalturaifi@gmail.com', the database name 'dbproject_students_ksu_soliman_fakeh_hosp', the copyright notice 'Copyright © 1999, 2024, Oracle and/or its affiliates.', and the version 'Oracle APEX 24.2.14'.

Result:

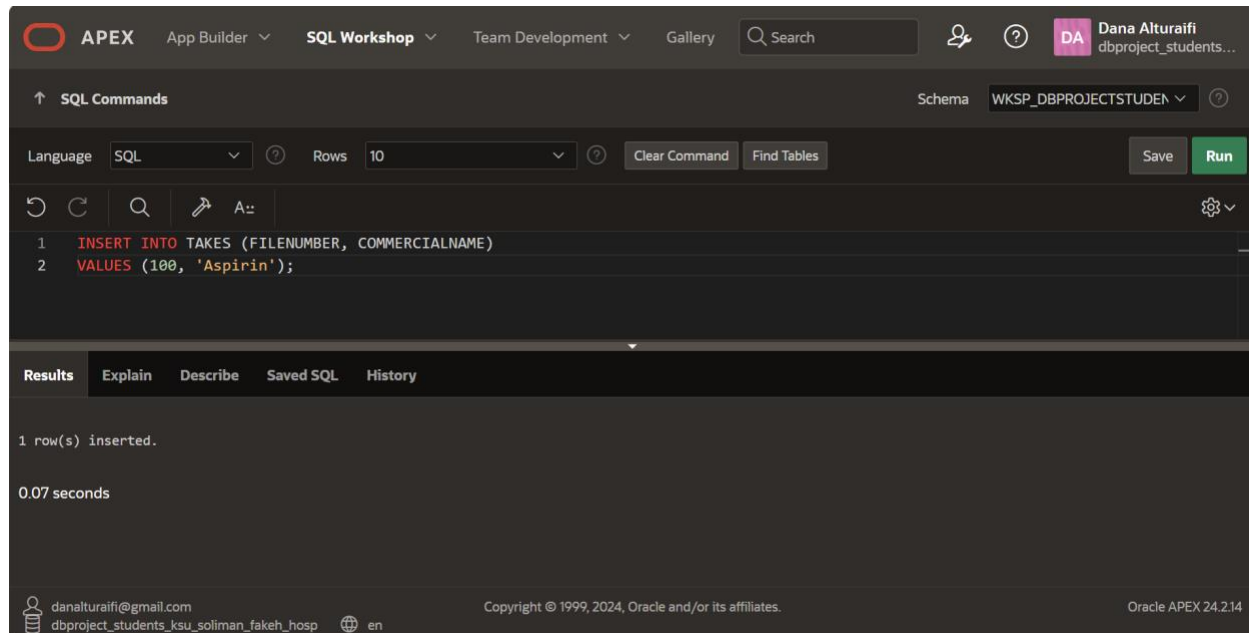


The screenshot shows the 'PRESCRIBES' table in the Oracle APEX interface. The table has two columns: 'DOCTORID' and 'COMMERCIALNAME'. The data is displayed in a table with four rows. The first row is highlighted with a blue border.

	DOCTORID	COMMERCIALNAME
	101	Aspirin
	101	Vitamin C
	201	Aspirin
	201	Vitamin C

Insert Table Takes:

Command :

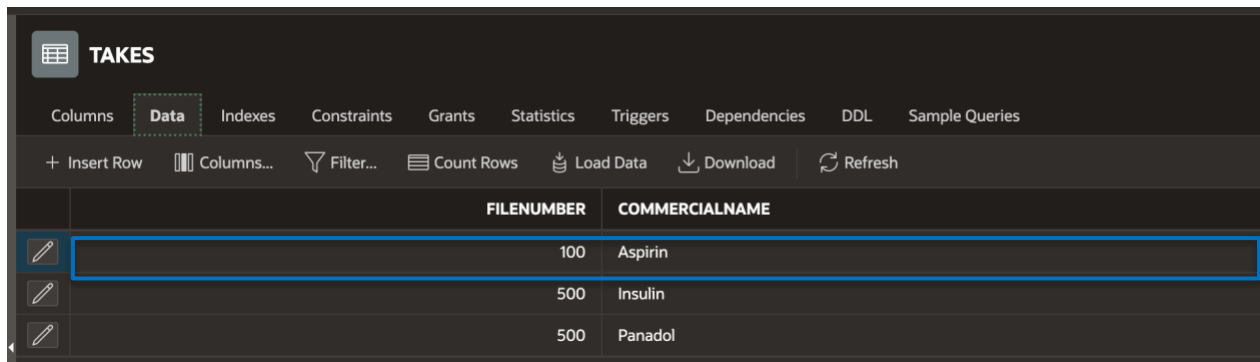


The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is logged in as 'Dana Altaifi' (dbproject_students...). The 'SQL Commands' tab is active, showing a schema of 'WKSP_DBPROJECTSTUDEN'. The language is set to 'SQL' and the number of rows is '10'. The command being executed is:

```
1 INSERT INTO TAKES (FILENUMBER, COMMERCIALNAME)
2 VALUES (100, 'Aspirin');
```

The 'Results' tab shows '1 row(s) inserted.' and '0.07 seconds'. The footer includes the user's email 'danalturaifi@gmail.com', the database name 'dbproject_students_ksu_soliman_fakeh_hosp', and the Oracle APEX version '24.2.14'.

Result:



The screenshot shows the 'TAKES' table in the Oracle APEX interface. The 'Data' tab is selected, displaying a table with two columns: 'FILENUMBER' and 'COMMERCIALNAME'. The table contains three rows of data. The first row is highlighted with a blue border.

	FILENUMBER	COMMERCIALNAME
	100	Aspirin
	500	Insulin
	500	Panadol

12.Data Queries:

1. View Daily Appointments

Lists scheduled appointments for Dr. Khalid (ID: 101) on May 10, 2026.

The screenshot shows the Oracle APEX SQL Workshop interface. The SQL Commands pane contains the following query:

```
1 SELECT * FROM APPOINTMENT
2 WHERE DOCTORID = 101 AND APP_DATE = TO_DATE('05/10/2026', 'MM/DD/YYYY');
```

The Results pane shows a single row of data:

APP_ID	APP_DATE	APP_TIME	STATUS	DEPT_CODE	FILENUMBER	DOCTORID
1	5/10/2026	1000	Confirmed	001	100	101

1 rows returned in 0.01 seconds

2. Find Doctors by Department

Displays a list of doctors in the Cardiology department.

The screenshot shows the Oracle APEX SQL Workshop interface. The SQL Commands pane contains the following query:

```
1 SELECT doc.NAME, doc.SPECIALIZATION, doc.ROOMNUM
2 FROM DOCTOR doc
3 JOIN DEPARTMENT dept ON doc.DEPT_CODE = dept.DEPT_CODE
4 WHERE dept.DEPT_NAME = 'Cardiology';
```

The Results pane shows three rows of data:

NAME	SPECIALIZATION	ROOMNUM
Dr. Nora	Orthopedic	102
Dr. Khalid	Cardiologist	101
Dr. Ahmed	General	103

3 rows returned in 0.03 seconds

3. Check Medicine Stock

Checks the current quantity of Panadol in the pharmacy.

The screenshot shows the APEX SQL Workshop interface. The SQL Commands pane contains the following query:

```
1 SELECT COMMERCIALNAME, QUANTITY
2 FROM MEDICINE
3 WHERE COMMERCIALNAME = 'Panadol';
```

The Results pane shows the query execution results:

COMMERCIALNAME	QUANTITY
Panadol	500

1 rows returned in 0.02 seconds. Download

4. View Patient History

Retrieves all appointments for patient Layan (FileNumber: 100).

The screenshot shows the APEX SQL Workshop interface. The SQL Commands pane contains the following query:

```
1 SELECT * FROM APPOINTMENT
2 WHERE FILENUMBER = 100;
```

The Results pane shows the query execution results:

APP_ID	APP_DATE	APP_TIME	STATUS	DEPT_CODE	FILENUMBER	DOCTORID
1	5/10/2026	1000	Confirmed	D01	100	101

1 rows returned in 0.02 seconds. Download

5. Check Medicine Price

Searches for Aspirin to retrieve its current price.

The screenshot shows the APEX SQL Workshop interface. The SQL command entered is:

```
1 SELECT COMMERCIALNAME, PRICE
2 FROM MEDICINE
3 WHERE COMMERCIALNAME = 'Aspirin';
```

The results are displayed in a table with two columns: COMMERCIALNAME and PRICE. The result shows Aspirin with a price of 25.

COMMERCIALNAME	PRICE
Aspirin	25

1 rows returned in 0.02 seconds. Download

6. List Department Details

Retrieves the location and contact number for the Cardiology department.

The screenshot shows the APEX SQL Workshop interface. The SQL command entered is:

```
1 SELECT DEPT_NAME, LOCATION, CONTACTNO
2 FROM DEPARTMENT
3 WHERE DEPT_NAME = 'Cardiology';
```

The results are displayed in a table with three columns: DEPT_NAME, LOCATION, and CONTACTNO. The result shows Cardiology with location Building A and contact number 112223333.

DEPT_NAME	LOCATION	CONTACTNO
Cardiology	Building A	112223333

1 rows returned in 0.02 seconds. Download

7. Filter Appointments by Status

Lists all appointments that are "Confirmed" for May 15, 2026.

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Dana Alturafi' are on the right. The 'SQL Commands' section shows a query in the 'SQL' language with 10 rows. The query is:

```
1 SELECT * FROM APPOINTMENT
2 WHERE STATUS = 'Confirmed' AND APP_DATE = TO_DATE('05/15/2026', 'MM/DD/YYYY');
```

The 'Results' tab is active, displaying a table with 7 columns: APP_ID, APP_DATE, APP_TIME, STATUS, DEPT_CODE, FILENUMBER, and DOCTORID. The table contains one row of data.

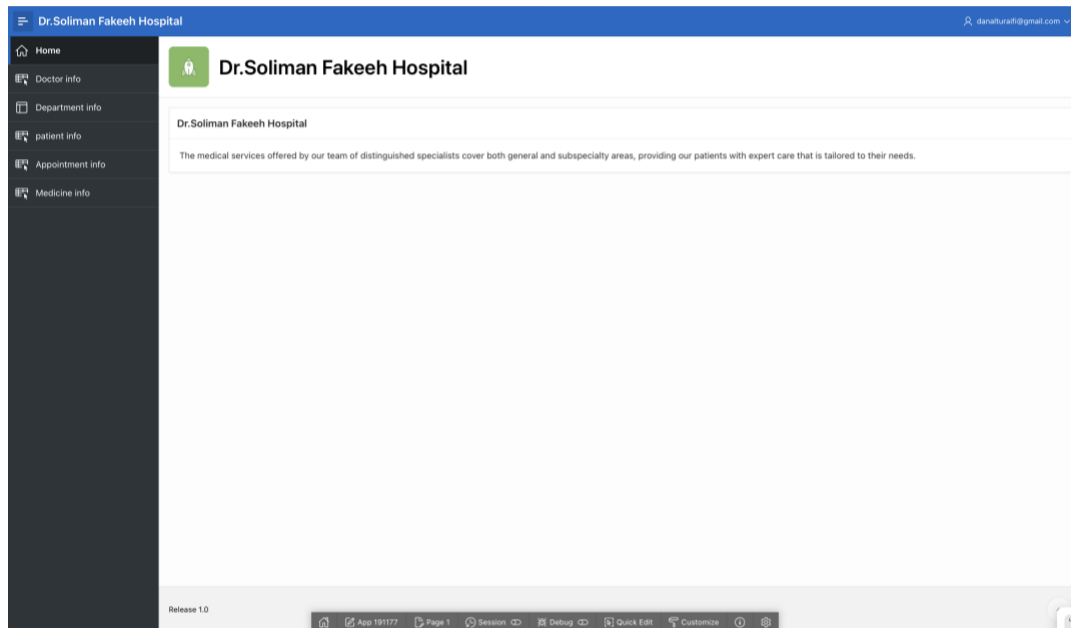
APP_ID	APP_DATE	APP_TIME	STATUS	DEPT_CODE	FILENUMBER	DOCTORID
2	5/15/2026	1100	Confirmed	D01	200	102

Below the table, it states '1 rows returned in 0.03 seconds' with a 'Download' link. The footer shows the user's email 'danalturafi@gmail.com', the database name 'dbproject_students_ksu_soliman_fakeeh_hosp', the copyright 'Copyright © 1999, 2024, Oracle and/or its affiliates.', and the version 'Oracle APEX 24.234'.

13.App Builder:

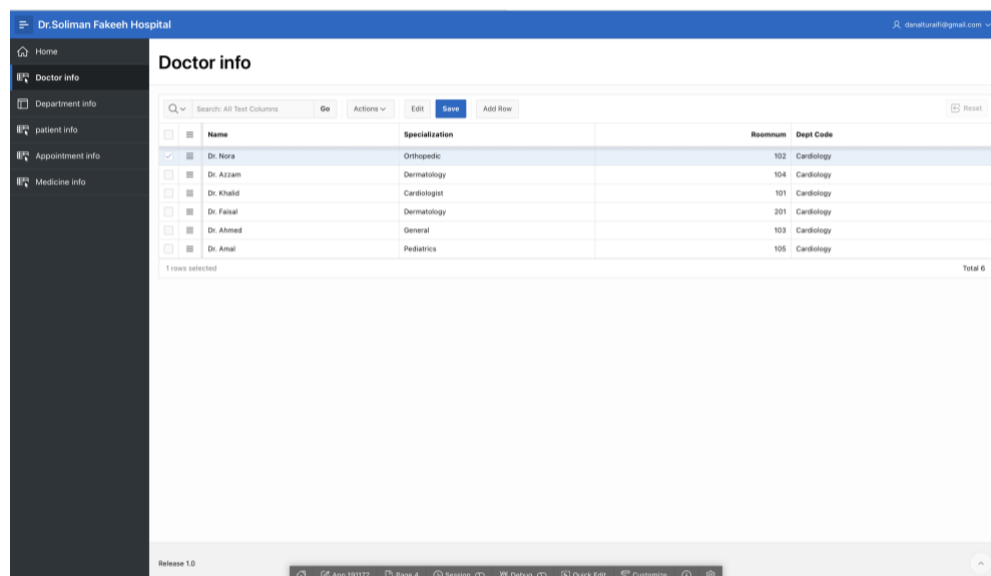
- **Home Page:**

This interface serves as the main entry point for the Dr. Soliman Fakeeh Hospital application. It includes a welcoming description of the hospital's services and a side navigation menu that provides easy access to all other sections of the system.



- **Interactive Grid:**

In the Doctor info, the features of adding, deleting, and editing have been enabled.



This screen displays the patient records, allowing for easy tracking of patient details.

Dr.Soliman Fakeeh Hospital

patient info

Search: All Text Columns Go Actions Reset

Nationalid	Dob	Gender	Name
1088776655	10/15/1995	M	Fahad
1033445566	11/10/2001	F	Noura
1099887766	5/20/2004	F	Layan
1077665544	1/12/1990	F	Sara
1122334499	3/10/1997	M	Sultan

1 rows selected Total 5

Release 1.0

App 191177 Page 2 Session Debug Quick Edit Customize

This interface shows the appointment scheduling system, linking patients with their respective doctors.

Dr.Soliman Fakeeh Hospital

Appointment info

Search: All Text Columns Go Actions Reset

App Date	App Time	Status	Dept Code	Filenumber	Doctorid
5/16/2026	1100	Confirmed	Cardiology	Fahad	Dr. Nora
5/10/2026	1000	Confirmed	Cardiology	Layan	Dr. Khalid
5/20/2026	1400	Confirmed	Radiology	Sultan	Dr. Fahad
5/20/2026	900	Confirmed	Radiology	Sultan	Dr. Fahad
6/10/2026	900	Confirmed	Cardiology	Sultan	Dr. Fahad
5/20/2026	930	Pending	Cardiology	Sara	Dr. Ahmed

1 rows selected Total 6

Release 1.0

App 191177 Page 8 Session Debug Quick Edit Customize

This section outlines the medicine inventory and details available in the hospital.

Price	Expiry Date	Quantity
45	11/20/2026	200
85.5	11/20/2028	40
120	10/6/2026	25
25	12/1/2027	100
15	5/1/2028	500
1 rows selected		Total 5

- **Master Detail:** The Department info table has been linked to the Doctor info table to display this data.

Dept Name	Location	Contactno
☐ Radiology	Building B	11224444
☒ Cardiology	Building A	112223333
☐ Laboratory	Building C	112225555
☐ Emergency	Building A	112226666
1 rows selected		Total 4

Name	Specialization	Roomnum
☒ Dr. Nora	Orthopedic	102
☐ Dr. Azzam	Dermatology	104
☐ Dr. Khalid	Cardiologist	101
☐ Dr. Faisal	Dermatology	201
☐ Dr. Ahmed	General	103
☐ Dr. Amal	Pediatrics	105
1 rows selected		Total 6

14. Work Distribution:

NAME	ID	Percentage	WORK
<i>Najla Almazyad</i>		20%	<i>Data Queries</i>
<i>Ghala alharbi</i>		20%	<i>App Builder</i>
<i>Dana Alturaifi</i>		20%	<i>Table Creation</i>
<i>Sara almazroa</i>		20%	<i>App Builder</i>
<i>Wajd Alotaibi</i>		20%	<i>Data insertion</i>

Table 4. Work Distribution